

Unit42

Sherlock Scenario

In this Sherlock, you will familiarize yourself with Sysmon logs and various useful EventIDs for identifying and analyzing malicious activities on a Windows system. Palo Alto's Unit42 recently conducted research on an UltraVNC campaign, wherein attackers utilized a backdoored version of UltraVNC to maintain access to systems. This lab is inspired by that campaign and guides participants through the initialaccess stage of the campaign.

Notes:

- Event ID 1: Process Creation/Execution. Includes process path, parent process path, and command-line arguments.
- Event ID 2: File Creation Time Changed. Includes the file making the change, the file to which the change is being made, tampered timestamp, and original timestamp.
- Event ID 3: Network Connection. Includes the process making the connection, destination IP Address, and port.
- Event ID 5: Process Termination. Includes the name of the process that was killed or terminated itself.
- Event ID 11: File Created. Includes the process creating the file, the file being created, and its full path.
- Event ID 22: DNS Query. Includes the process querying the domain, the target domain name, and the IP Addresses they resolve to.

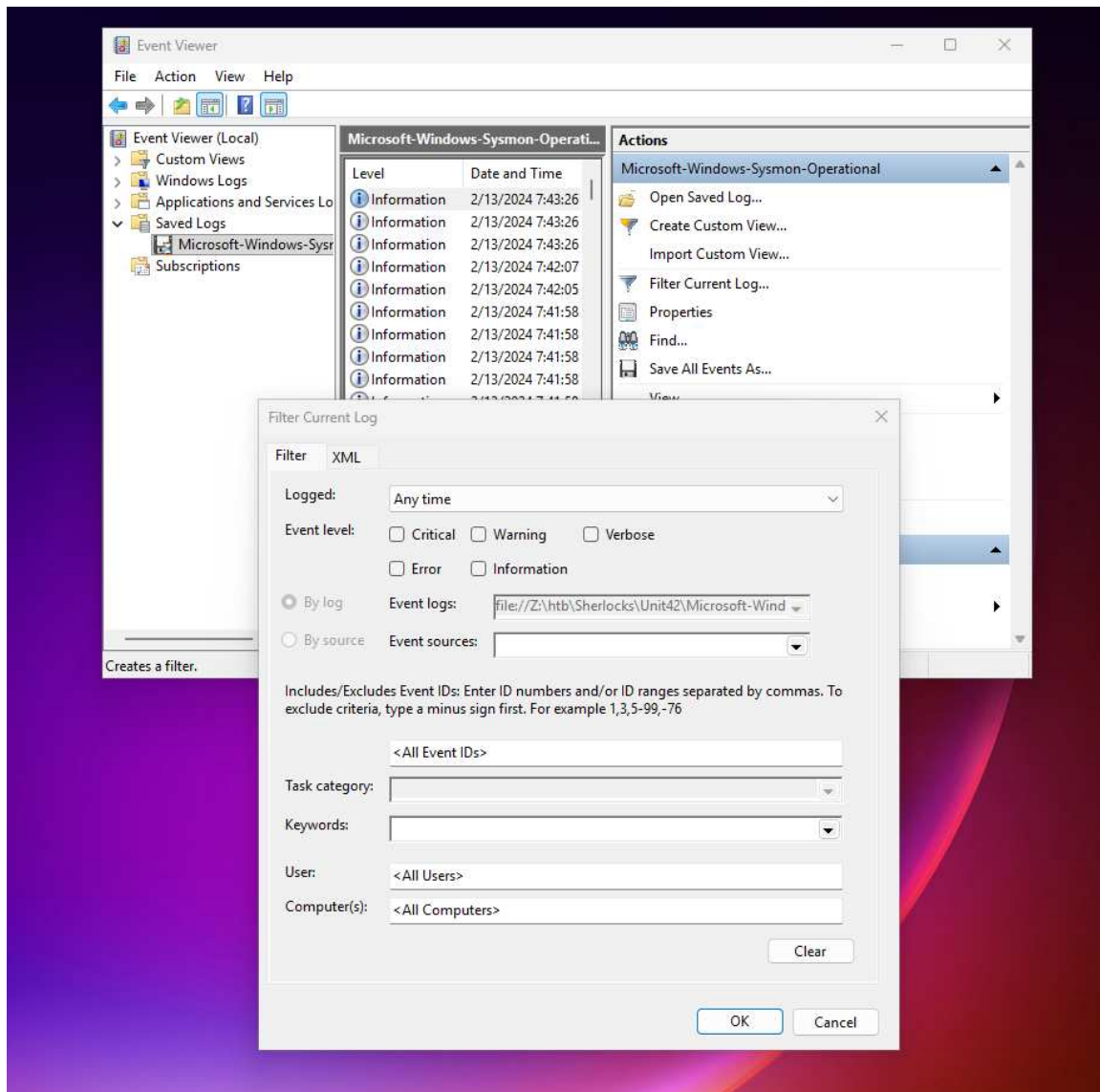
Knowing this before entering the investigation help you in further future analyzes

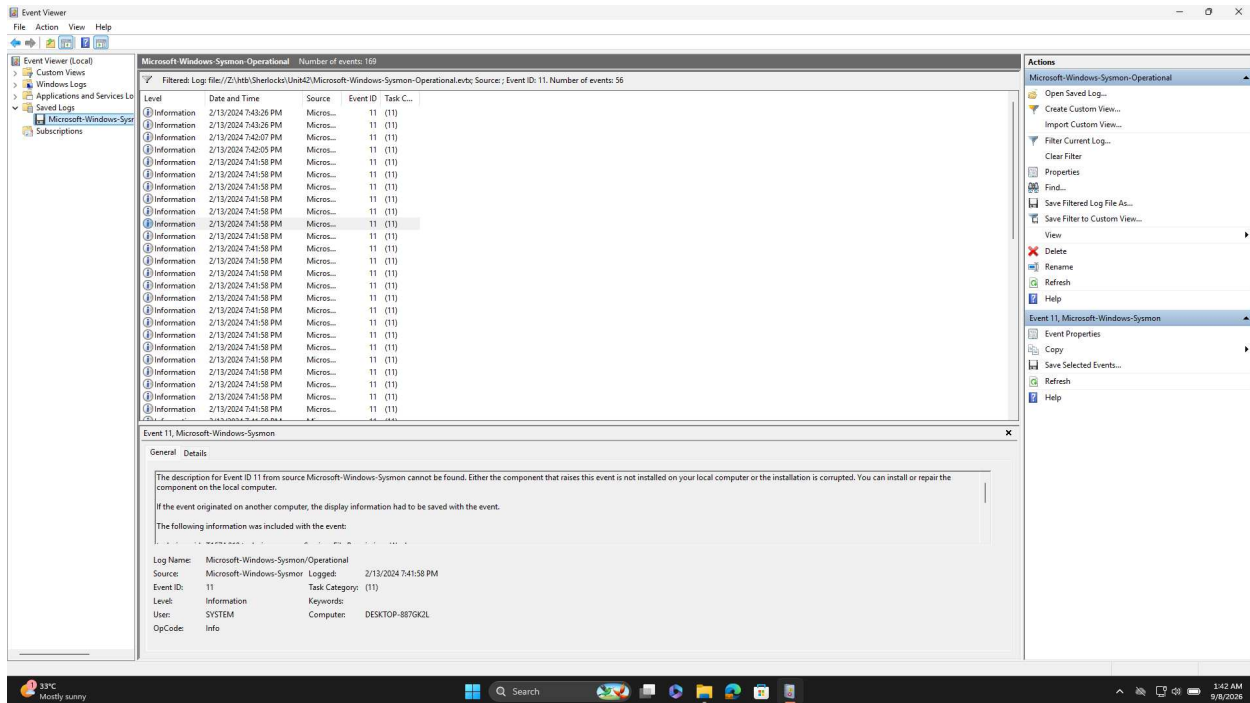
→ Download the zip - you will get the Microsoft-Windows-Sysmon-Operational - Event log file

No need to convert the evtx to csv - if you had windows you can just view them in Event Viewer -it windows service provided to analyse the logs and track the log activities

1.How many Event logs are there with Event ID 11?

Click the Filter current logs under actions it opens a tab in the place of <All Event ID> mention 11





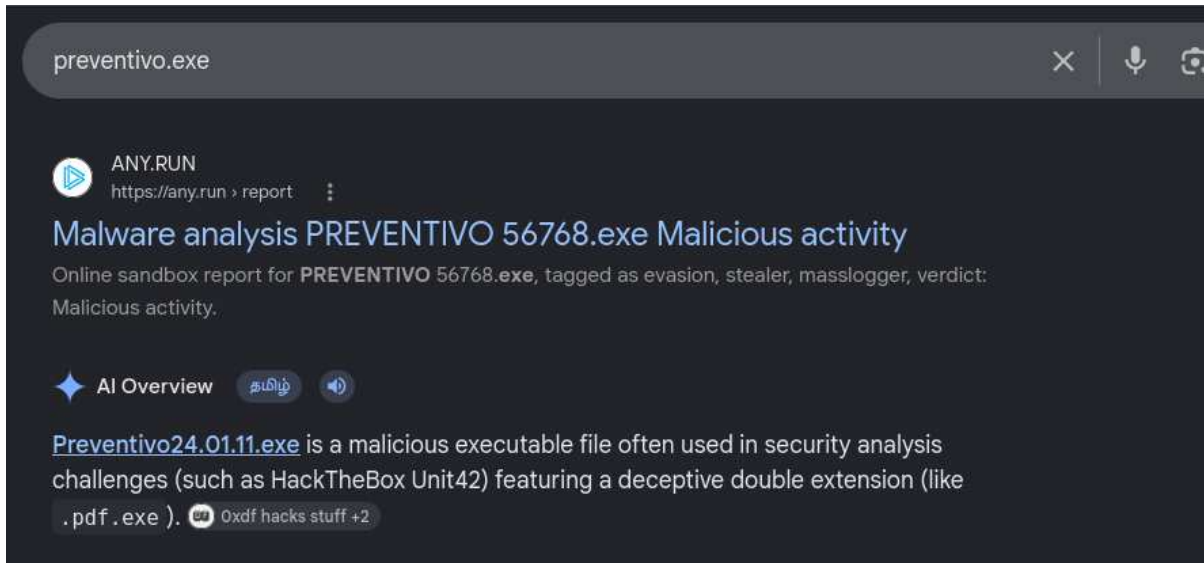
56

2. Whenever a process is created in memory, an event with Event ID 1 is recorded with details such as command line, hashes, process path, parent process path, etc. This information is very useful for an analyst because it allows us to see all programs executed on a system, which means we can spot any malicious processes being executed. What is the malicious process that infected the victim's system?

NOTE the question : Whenever a process is created in memory, an event with Event ID 1

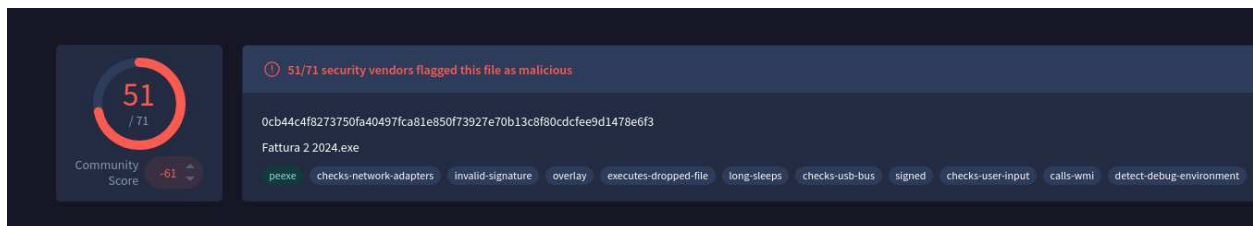
Filter out Event ID 1

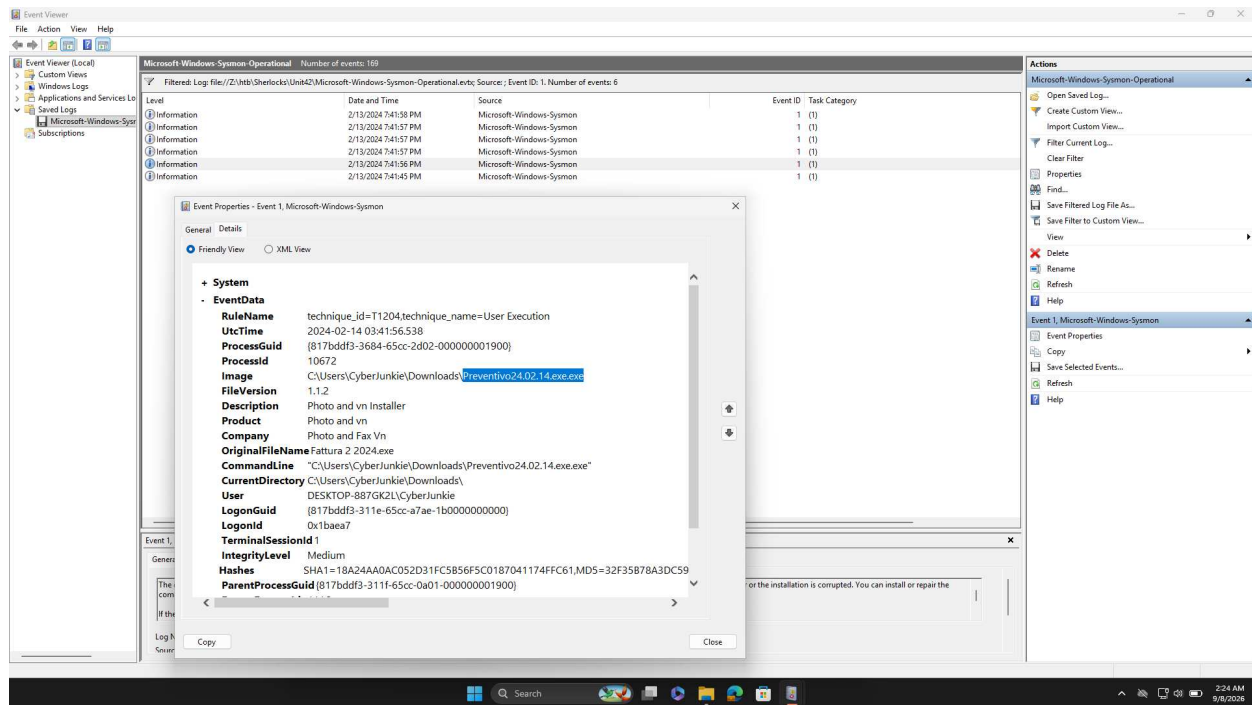
Search for this .exe file in Internet you will find it is malicious



Other way to confirm is through Virus Total

MD5=32F35B78A3DC5949CE3C99F2981DEF6B

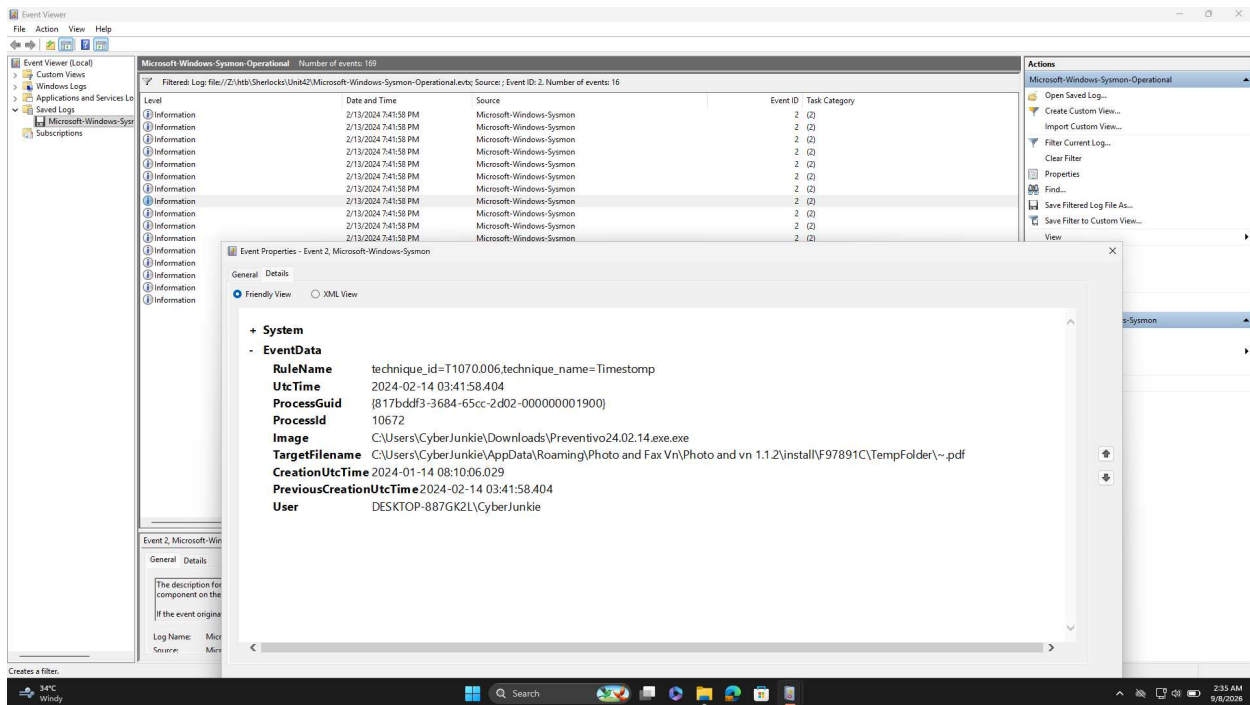




C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe

3.Which Cloud drive was used to distribute the malware?

Event ID 22 can be used to look for any DNS Queries made by the system

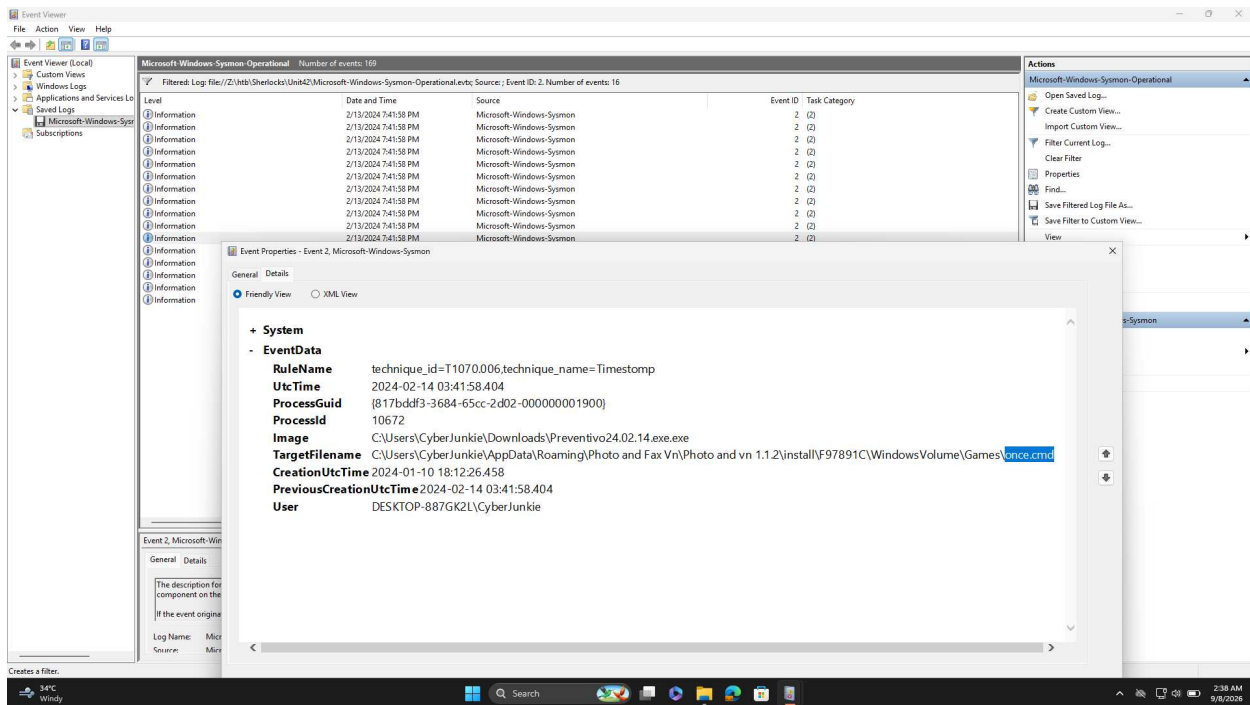


Note: Look at the previous Creation UtcTime this is the actual "original timestamp" the Creation Timestamp is the manipulated/changed timestamp

2024-01-14 08:10:06

5. The malicious file dropped a few files on disk. Where was "once.cmd" created on disk? Please answer with the full path along with the filename.

Just from the same place of above question look all the timestamps are same looks like all files are dropped at the same time. In the left side we see the above and below icon click and see till you come through the path that contains "
\\once.cmd"



C:\Users\CyberJunkie\AppData\Roaming\Photo and Fax Vn\Photo and vn 1.1.2\install\F97891C\WindowsVolume\Games\once.cmd

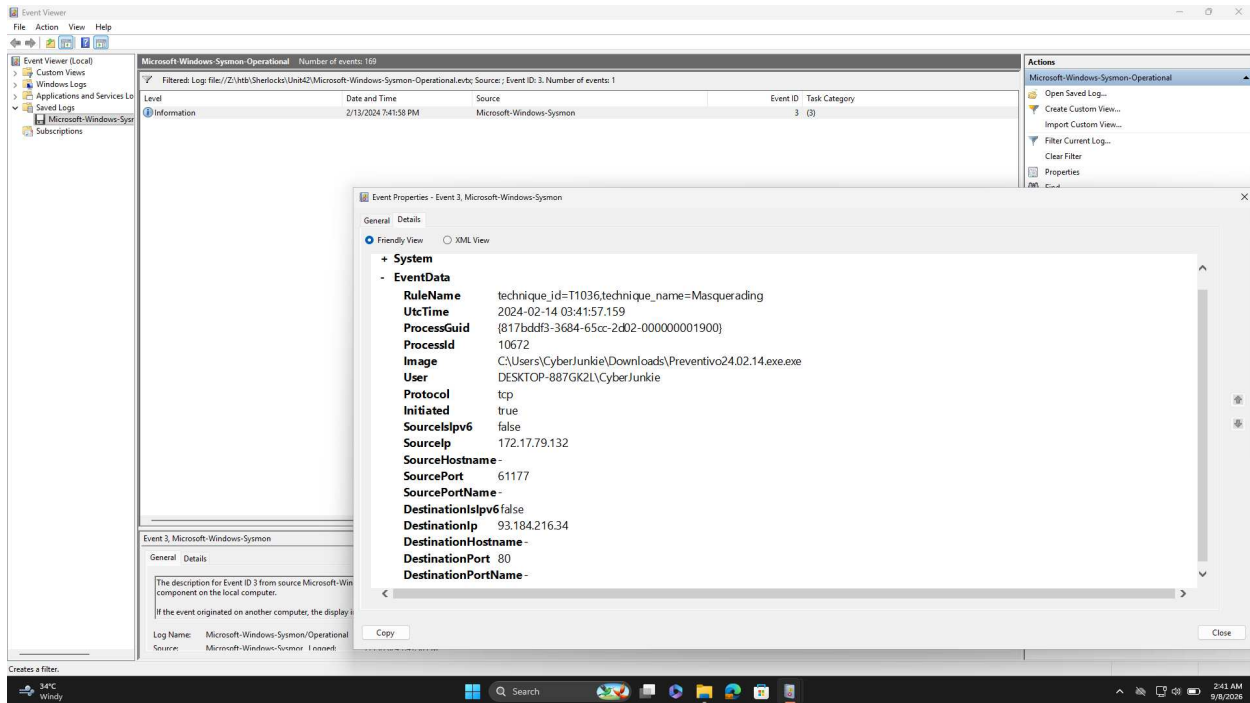
6.The malicious file attempted to reach a dummy domain, most likely to check the internet connection status. What domain name did it try to connect to?

Look back through the event ID 22 search the DNS queries made for the test purpose he queried www.example.com

www.example.com

7.Which IP address did the malicious process try to reach out to?

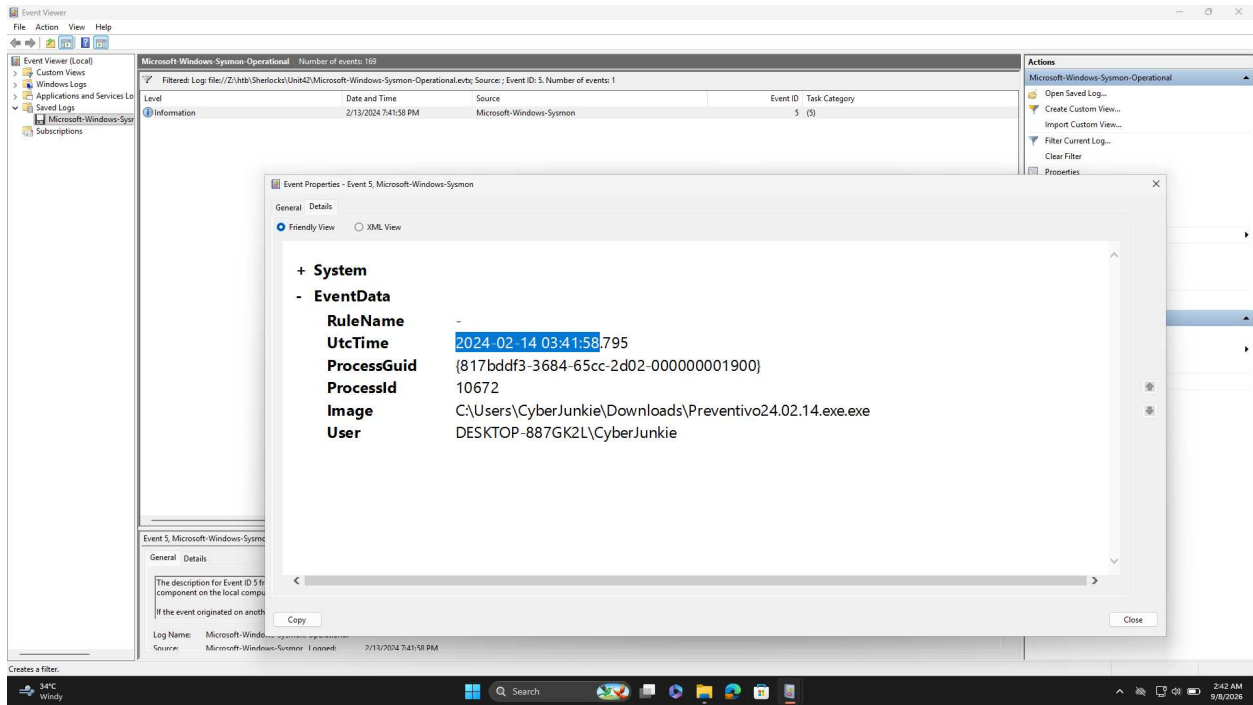
Event ID 3 - It records the IP address, port, and the process trying to make the connection.



93.184.216.34

8.The malicious process terminated itself after infecting the PC with a backdoored variant of UltraVNC. When did the process terminate itself?

Filter for Event ID 5 and look for the Image name, which should be the malicious process.



2024-02-14 03:41:58